

ECEN4233 High Speed Computer Arithmetic Project

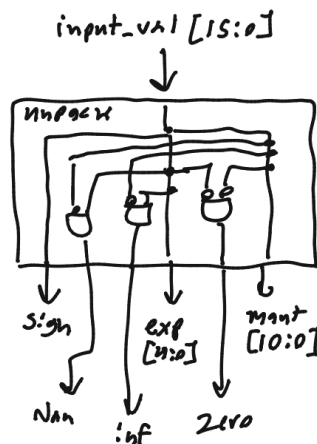
Skylar Araujo

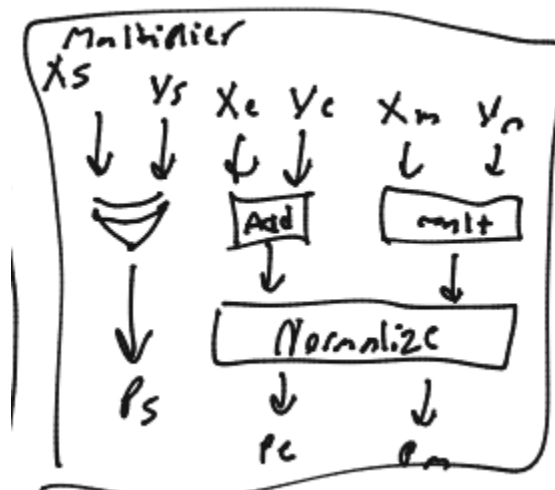
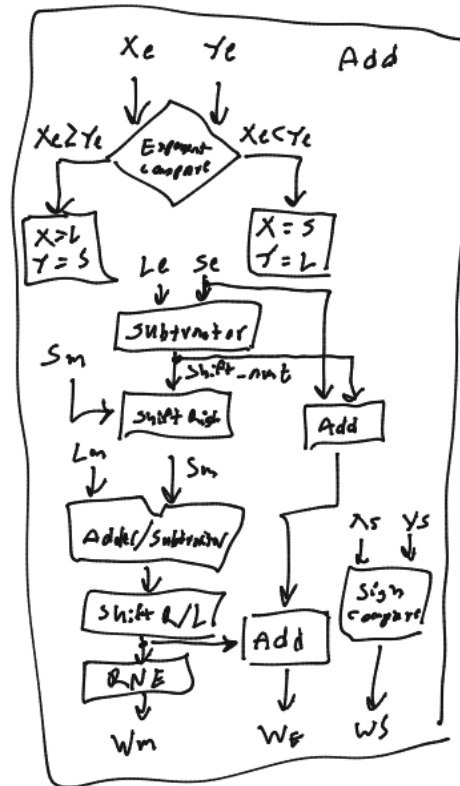
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Introduction

While the primary focus of this project was on the FMA, I was unable to finish the project and produce a fully working design. I allowed my workload to grow to an unmanageable degree and had to prioritize other classes and responsibilities at the expense of this one. I did not give myself nearly enough time for this project and completely exhausted myself with other classes and responsibilities. The majority of my efforts went into understanding and designing the FADD procedure.

Block Diagrams





Architecture

A Fused Multiply-Adder (FMA) is a floating-point architecture that performs a multiplication and an addition in a single, combined operation, all in one rounding step:

$$F = A * B + C$$

Unlike performing a multiply followed by a separate add—which would require two normalizations and two roundings—the FMA architecture keeps the full-precision product

internally (typically using extended mantissa bits), aligns it with the addend, performs the addition, then normalizes and rounds only once. This greatly improves both numerical accuracy and performance. Architecturally, an FMA typically contains a high-precision multiplier, a wide carry-save accumulation stage, and a final normalization/rounding unit. FMAs are essential in modern CPUs and GPUs because they reduce rounding error, reduce energy/latency compared to separate MUL + ADD units, and enable high-throughput operations common in DSP, scientific computing, and machine learning.

Optimization

A fused multiply-adder (FMA) is optimized by performing the multiply and add in a single fused operation with only one rounding, improving both speed and accuracy. It uses carry-save accumulation to combine the product and addend efficiently, keeps extra internal precision to reduce alignment and rounding error, and shares a single normalization/rounding unit instead of two separate ones. Early detection of special cases and pipelining further reduce latency and hardware cost, making the FMA both fast and numerically stable.

Test Result

User Defined Tests

I was unable to thoroughly test any of the units I designed. I created simple test benches with values and answers I calculated by hand.

Project Management

Time Management

Unfortunately, poor time management has greatly impacted my performance on this project. My greatest challenge this semester has been balancing classwork, club responsibilities, my part-time job, and unforeseen personal circumstances. As a result, I failed to keep up with the suggested schedule for completing the project and fell far too behind. By the time I got around to working on the project, I realized there was simply far too much work to be done in the limited time I've given myself to work on. I wish I had heeded your warnings and started early. I recognize that I should have adjusted my workload earlier before allowing it to spiral out of control.

Reflections on Lessons for Future Projects

This project has served as a dire wake-up call. I entered this semester far too ambitious and overloaded myself with more responsibilities than I could manage. As deadlines approached, the gap between where I was and where I needed to be became overwhelming. What disappoints me the most is that I really enjoyed the material in this class. I regret not giving myself the time I

needed to fully engage with the project. Moving forward, I plan to significantly cut back on the amount of work I take on at once. Rather than taking on as much as I can at once, I will be more intentional with my commitments and ensure that I do not sacrifice the quality of my work, or my well-being, by overwhelming myself. I stretched myself too thin and have seen the consequences in doing so. This is not a mistake I want to commit again. I apologize for not giving this class the time it deserved. If I could retake it, I most certainly would.

Conclusion

The design and implementation of the fused multiply-add module have proved incredibly challenging for me. While I failed to give this project the time it needed to get the most out of it, it would be wrong to say I learned nothing. Even though I was only able to complete the floating-point addition and multiplication units, working through these components has significantly improved my understanding of how floating-point arithmetic is implemented and the process behind it.